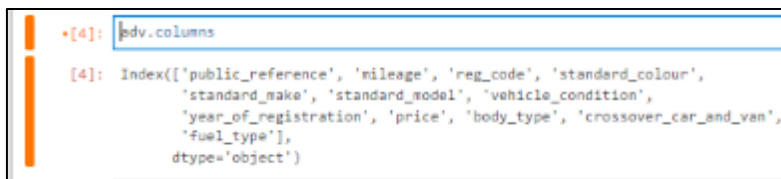


1. Data Understanding and Exploration -

The adverts.csv file is a dataset storing details regarding car listings that have been advertised for sale on the Autotrader website, along with relevant information relating to the listed car. Once the dataset was successfully imported via Pandas into the Jupyter notebooks workspace, a data inspection was performed.

1.1 Meaning and type features -

It was necessary to ascertain the relevant features present in the dataset to establish the information they convey. **Figure 1** shows the listed features present in the adverts.csv dataset.



```
[4]: pdv.columns  
[4]: Index(['public_reference', 'mileage', 'reg_code', 'standard_colour',  
         'standard_make', 'standard_model', 'vehicle_condition',  
         'year_of_registration', 'price', 'body_type', 'crossover_car_and_van',  
         'fuel_type'],  
        dtype='object')
```

Figure 1 – Features Present In The Dataset

Reg_code, short hand for registration code is a single letter ranging from A to Y situated at the suffix of the registration plate for cars registered from 1963-1983, or as the prefix for cars registered from 1983-2001. For cars registered from 2001 to the present day, the system was replaced with two numbers typically contained within the middle of a vehicles registration plate. These codes, which are used to indicate the vehicle's first registration date, is updated biannually in March and September and has been since March 1999 (prior to this was updated every year). This convention makes it possible to determine a car's age within a six-month period simply from its licence plate, or within a year if registered pre March 1999. Due to codes being reused and the lack of information as to whether a suffix or prefix is captured in the dataset, using letters alone as a means of identifying the year registered can yield unexpected outputs as each code can potentially correspond to a choice of three years, one being correct. Column Year_of_Registration would instead be more appropriate.

While Standard_make is the company or brand that manufactures the car e.g. BMW, Standard_model is the specific design or version of a vehicle produced by the manufacturer e.g. BMW **1 series**. This has been identified as the most physically unique feature in the dataset, and effectively offers a greater level of granularity than Standard_make. The model can typically indicate the body shape, features present in the car, as well as a rough indication as to when the car manufactured.

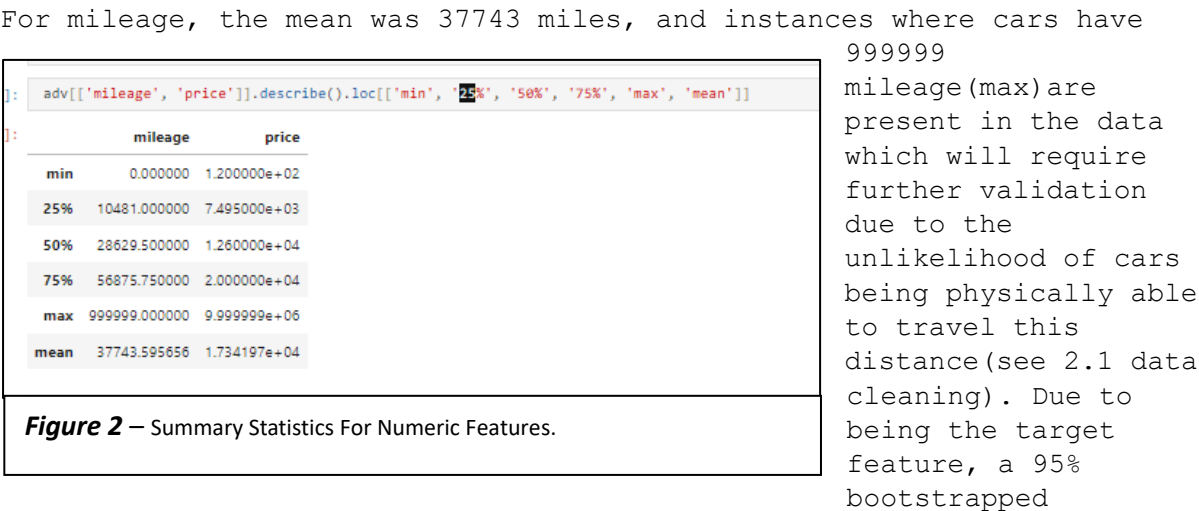
Vehicle_condition denotes the state of the car, with the two possible options indicating whether it has been previously owned by another driver (USED) or not (NEW). There are a number instances where mileage has been accumulated on the odometer (in some cases up to 50 miles) which can be attributed to reasons such as quality

control purposes, test drives, transportation to dealership and more) while the vehicle condition is NEW, therefore this feature isn't a measure of whether the car has been driven before, but a classification of ownership.

Price is the advertised cost of the car determined by the seller, based on their own individual valuation. As Autotrader is a British company, the currency has been assumed to be measured Great British Pounds (£). Price is feature of interest in the report, and investigating the relationship between price and the other features in the dataset will serve as the foundation to evaluate market dynamics.

1.2. Analysis of Distributions –

To further come to terms with the data it was necessary to investigate the relative distributions of each of the features present in the dataset. **Figure 2** provides the summary statistics for the numeric features (year not included as is ordinal)



confidence interval for the mean of price was obtained to observe the effects of extreme values, with upper and lower limits corresponding to £17131 And £17398 respectively, and the actual mean for the entire data falling within this range at £17256. The median valuation was £12,600, and 75% of cars cost less than £20,000. However, there is a wide distribution of values ranging from £120(min)to £9999999(max).

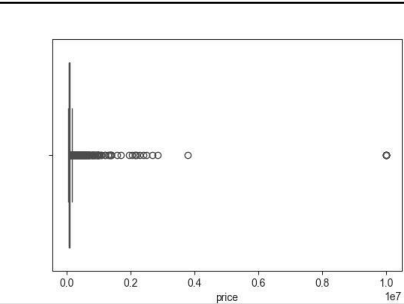
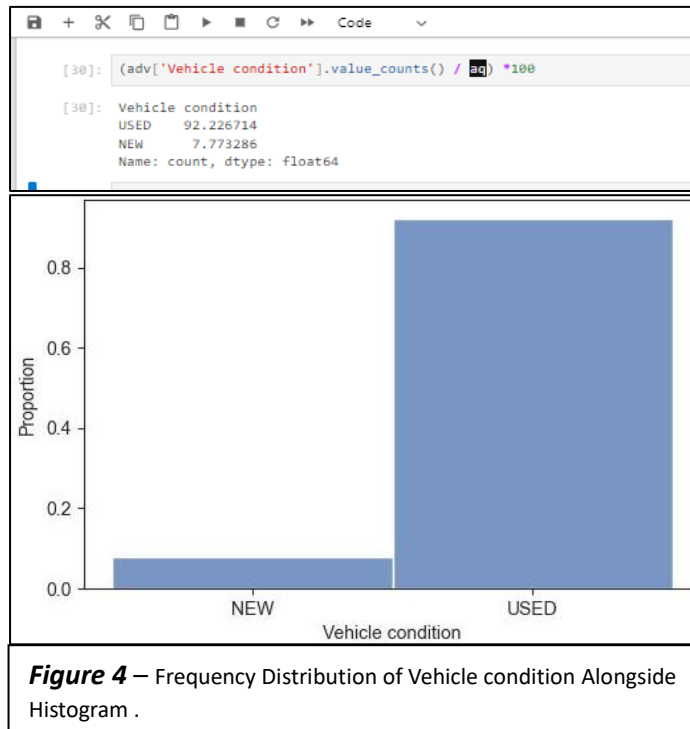


Figure 3 - Boxplot of Price.

To clarify whether these are erroneous values, a boxplot of price (**Figure 3**) was produced; The boxplot itself is incoherent, however it does allow for further understanding of the impact these extreme values pose for subsequent analysis. While there does appear to be a fair number of outliers costing several magnitudes greater than that observed for the bulk of the distribution, these can be attributed to rare, vintage or supercars due to their lack of occurrences, with the bulk of the

dataset containing cars that are financially accessible to a broader spectrum of clientele.

The frequency distribution of vehicle condition provides insight into the type of market (NEW or USED) the dataset mostly captures, in turn directing the scope of the subsequent analysis.



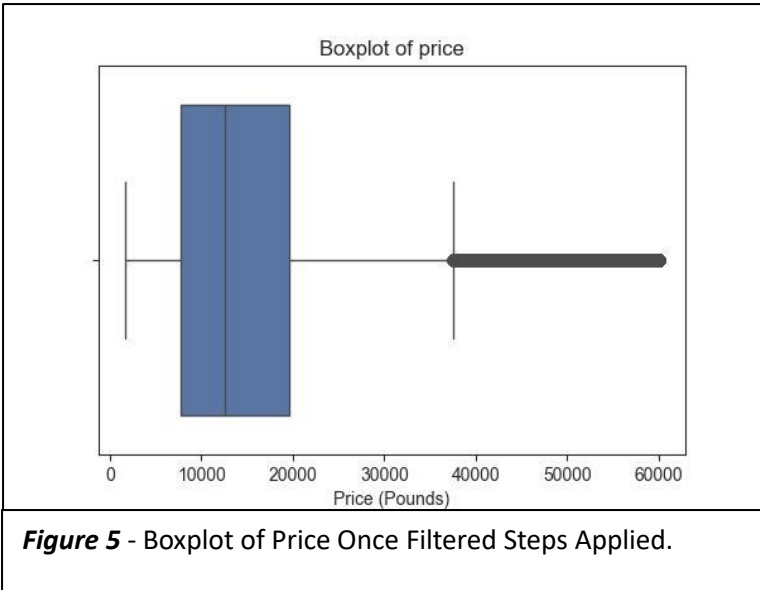
From **figure 4**, the disparity within vehicle condition leans heavily towards used cars, accounting 92.2% of all car listings. Depending on the relationship observed and its influence on price, executing a bivariate analysis when examining the data may be most appropriate.

2. Data Pre-Processing -

2.1 Data Cleaning (e.g., dealing with incorrect values, outliers) -

To overcome the issues encountered during the data exploration and to ensure integrity, reliability, and usability of data for analysis it was necessary to deal with outliers and incorrect values. The most expensive cars present in the dataset, while appearing ludicrously priced were confirmed to be appropriate given the model of the car. While not erroneous, they are not representative of the bulk of the dataset, being out of consideration for the vast majority of consumers exemplified by the little representation within the data. On the other hand, extremely cheap cars will hinder the accuracy of the analysis as there is likely be an unmeasured variable rationalising the low price. This may be unknown damage unspecified in the dataset, for example a car having cat s/n write off status, an illegal marker or the seller simply wanting to sell swiftly thus presenting as a bargain. It is therefore safe to conclude that the integrity of these values is questionable as an accurate representation of price given the parameters captured in the dataset. However, given the prevalence of the cheap car market i.e. for first time drivers who are likely to damage a car, it was important that the data captures this segment of the market.

The strategy of eliminating both the top 2.5% and bottom 2.5% of cars with respect to price was implemented having been classified as outliers, narrowing the data to be examined within the price range of £1695.0 to £59995.0.

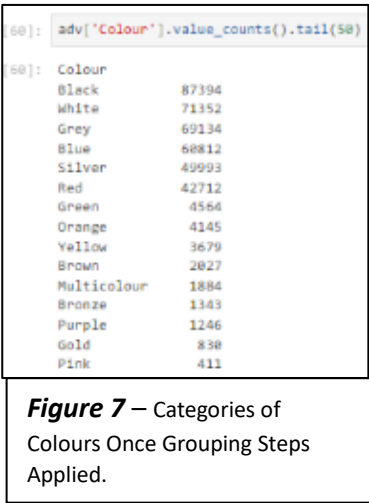
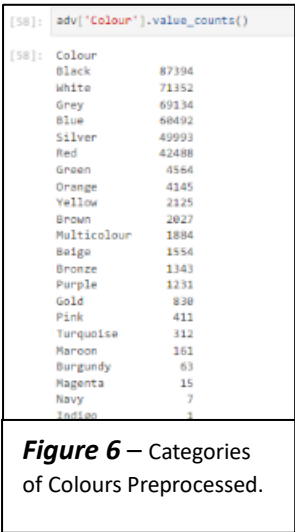


Similar steps were applied with the mileage, instead using a one-sided approach to remove the upper 0.5% of values. Cars typically become more uneconomical to repair upon approaching 200,000 miles, so the merchantable quality of cars when approaching this figure, regardless of potential other enticing features (particularly price) reduces significantly. Meanwhile, cars with very low mileage simply may not have been driven so it would be inappropriate to

treat them as extremes of values and therefore the data for subsequent analysis was narrowed to between 0-150,000 miles.

Inaccurate and unexpected values, such as cars listed as having been registered in the year 999, were discovered within the year of registration column. Once confirmed these were not more primitive forms of transport and consequently determined to be erroneous (assumably down to input error), it was necessary to remove these incorrect values. Upon further investigation, there were fewer than 30 instances per year where the registration date was pre 1986. Therefore, 1986 was determined as the oldest year to be included for subsequent analysis.

Specific shades instead of the more broader colours groups were identified while investigating the distribution of standard colours, for example indigo as a shade of blue and burgundy as a shade of red. It was necessary to map these shades to the more encompassing colour classifications so there is a consistent level of granularity for this feature throughout the dataset.



2.2 Feature engineering -

It was possible to derive new informative features from those already present. The latest vehicle registration date was from 2020, so under the context that this is the most recent date in which the data was collected, recorded and assembled the feature "Milesperyear" (**Figure 8**) was produced, providing the average number of miles driven per year per car. This feature provides information as to the driving behaviours and how frequently and extensively a vehicle has been driven, as well as an indication towards the

```
[21]: adv['milesperyear'] = adv['mileage'] / (2020 - adv['year_of_registration']).replace(0, 1)
```

Figure 8 – Code Implemented to Engineer New Feature "Milesperyear"

overall picture of the vehicle's utilisation, whether that be more

leisurely/recreational use or regular lengthy commutes.

To determine the average price per car make depending on the condition of the vehicle, cars of the same make and condition were grouped together and introduced into the dataset (**Figure 9**). The identical procedure was followed, except the equivalent columns were produced using the model of the car. Comparing and contrasting the new and used car markets, and ultimately increasing the level of granularity from make to model will provide us with useful information regarding value retention between respective categories. Note. Respective columns won't be populated for categories that don't contain any instances of either new or used data.

```
adv["AvgMakeUsed"] = adv[adv['Vehicle condition'] == 0].groupby(['Make'])['Price'].transform('mean')
adv["AvgMakeUsed"] = adv["AvgMakeUsed"].fillna(adv.groupby("Make")["AvgMakeUsed"].transform("mean"))
```

Figure 9 – Code Implemented to Engineer New Feature "AvgMakelUsed" (Average Model Price Used).

To be able to observe visually which car models retain the most amount of value, the dataframe "Carmodel" was produced (**Figure 10**), containing each make of car with their corresponding average price when new and used where applicable, along with the average mileage. The difference between the average prices was then calculated in the column differences.

```
[276]: Carmodel.sample(5)
```

	Model	AvgModelUsed	AvgModelNew	AvgMileage	Differences
693	Rapid Spaceback	8262.748815	NaN	36798.488152	NaN
439	Giulia	27137.132231	39691.93913	12328.755618	12554.806899
606	Outlander	18313.991137	35086.54902	33707.922156	16772.557882
465	HiAce	6208.500000	NaN	67741.000000	NaN
964	ix20	7845.760369	15253.00000	30338.331019	7407.239631

Figure 10 – Snippet of Dataframe Carmake.

The typecasting of vehicle condition to a numeric data type was performed, specifically to binary form with 0 and 1 representing used and new cars respectively to numerically assess the impact as a feature.

2.3 Subsetting -

The discretization of cars based on price to generate a new column price "Category" was executed (**Figure 11**), placing cars into 1 of 5 bins to further characterise. Although bins are usually separated into

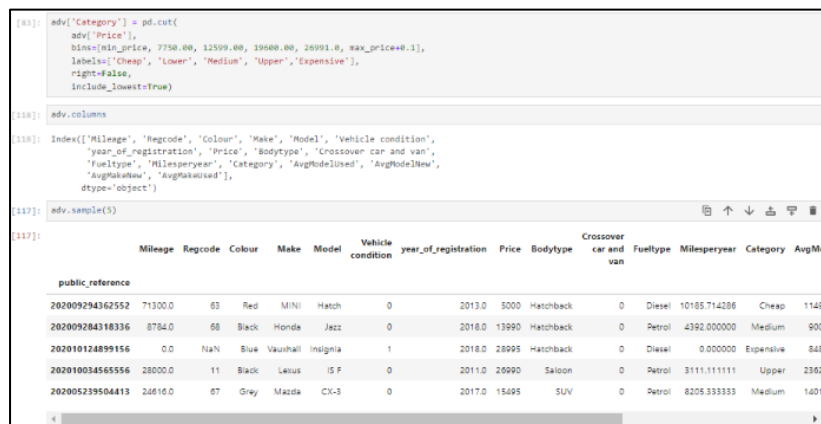


Figure 11 – Code Applied to Discretise Price, Resultant Dataset Displaying New Introduced Column.

equal fragments, a custom classification model was applied and the bins were determined using the summary statistics. The bins Cheap, Lower, Medium, Upper and Expensive correspond to the percentile ranges 0-25th, 25th-50th, 50th-75th, 75th-87.5th and 87.5th-100th respectively with regards to price

to be used for a multivariate analysis.

In order to validate the modifications applied during the pre-processing steps row sampling was instituted. **Figure 12** shows the appearance of the pre-processed data frame after the initially importation into Jupyter Notebooks, in comparison once treated as detailed throughout in 2. Data Preprocessing (**Figure 11**).

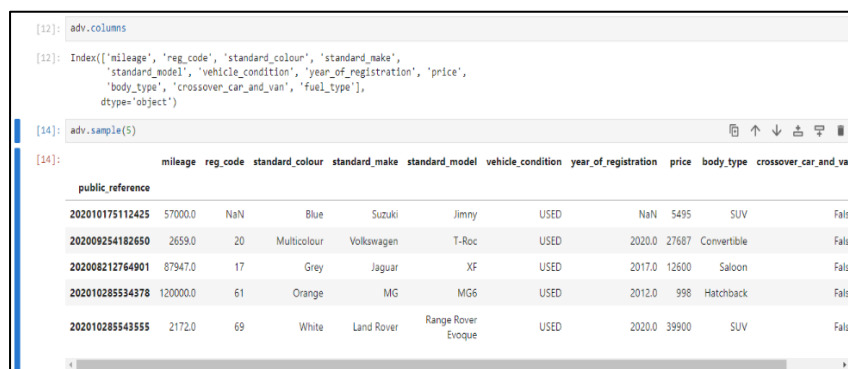
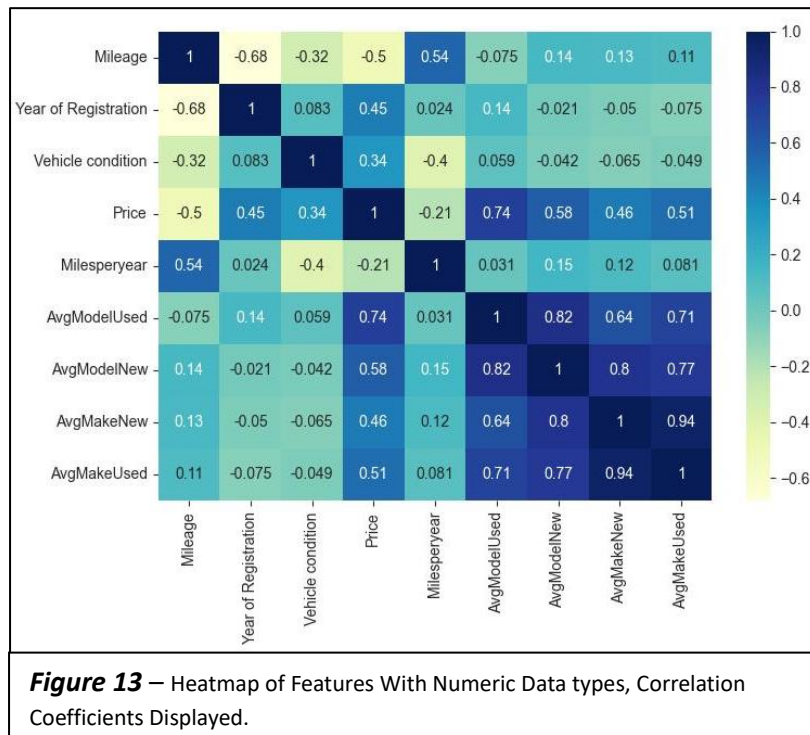


Figure 12 – Five Random Sample Instances of Data (Pre-processed).

The new calculated columns are listed and have been incorporated into the dataset with the corresponding values, as well as general housekeeping steps for further clarity.

3. Analysis of Associations and Group Differences -

3.1 Quantitative-quantitative -



To quantify the relationships between numeric features, a heatmap was produced as seen in **Figure 13**. The feature that is most strongly correlated with price is the derived AvgModelUsed with a positive correlation coefficient of 0.74. This is attributed to the fact that the model of the car is the most unique physical identifier as re-emphasised throughout this report and while not completely independent of price,

is the greatest indication. AvgModelNew, while having the second largest correlation coefficient (0.58) has less of an influence in comparison due to the heavy skew in the distribution of vehicle condition towards used cars. Similarly, The AvgMakeUsed exhibits a moderate correlation with price with a correlation coefficient of 0.51, indicating that the prestige and reputation of the car manufacturer has a significant influence on price. Potentially if it were more widely known that manufacturers share parts (for example, sports car Pagani Zonda using the same engine as pedestrian Mercedes-Benz S600) this observed correlation may be lower.

Mileage has a moderate negative correlation with price (-0.5), attributed to car parts being mechanical so the further the distance travelled the greater the likelihood of the depreciation of the moving parts required. Albeit wear and tear can be dependent on how carefully the vehicle has been driven and its upkeep, but collecting the information as to the main reason the car was utilised to provide further information towards this would be a useful metric to capture. However, because of the sheer number of components involved in making a car run, the vast majority will eventually become uneconomical to repair with increasing mileage. Depreciation of parts also leads to a reduction in performance, further explaining the observed relationship. The intensity in which a car has been driven has noticeably less of a bearing on price as observed with its correlation with milesperyear (-0.21).

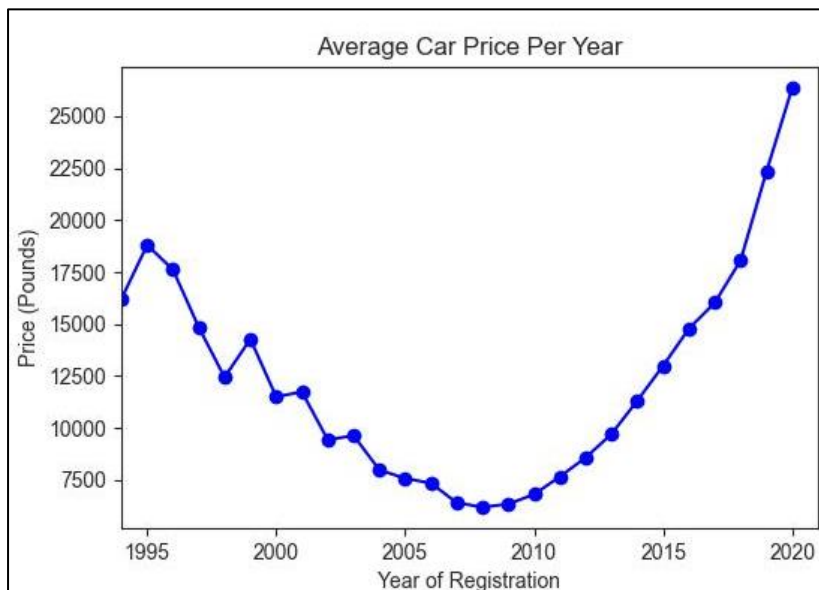


Figure 16 – Average Price of Cars Per Year, 1995-2020.

From 1995 to 2009 the average price of cars has been generally decreasing, with exceptions between the years 1997-1998, 2000-2001 and 2002-2003 but the overall relationship holds true. Older cars that still appeal to consumers today and therefore listed on Autotrader will be specifically sought after for a certain reason, may that be rarity, driving experience or unique

features/components no longer present in modern cars. Furthermore, changes in legislature in 2000 saw the banning of leaded petrol, which was the predominant fuel type at the time. This meant cars running on leaded petrol had to either be scrapped or converted to run on unleaded, with an exemption to the ban existing for owners of classic cars. As the level of granularity within fuel types doesn't allow us to explicitly see which type of petrol (leaded or unleaded) these cars take, it can be assumed the proportion of classic cars to standard for cars registered pre 2000 is expected to be increase as the year registered decreases.

This can be observed for the average price of cars registered from 2000-2008 but to a lesser extent as cars were no longer produced to run on leaded petrol. Prices for cars registered around the early 2000s are typically cheaper than more modern versions because they have had sufficient time to accumulate a greater number of miles (correlation coefficient between mileage/year of registration is moderate/strong 0.68, **Figure 15**) thus affecting price. This will have less significance on price the older cars become attributed to the increase in prevalence of rare/vintage models as mentioned.

From 2008-2009 however this trend changed, and since then prices have been increasing year on year. Worldwide, in 2007-2008 the global financial crisis broke, and in the third quarter of 2008 the UK economy had moved into a technical recession which turned out to be the deepest since the second world war. The weakening of the pound against other global currencies resulted in cars produced since 2009 costing more in the UK market than previously, with foreign imports accounting for the majority of cars driven in the UK.

The notable increase in average car price since 2018 can be understood given that a new car loses value as soon as it is purchased and driven off the forecourt and by the end of the first year although varied, can lose around 40% of its value. In combination with the increasing

proportions of new cars since 2018(9.1% in 2018, 15.0% in 2020)as well as the shrinkage in the market for cars registered post 2018(73139 in 2017, 41484, 42703, 31747 in 2018, 2019 and 2020 respectively). Furthermore, the introduction of more expensive technology attributed to the shift in fuel types (see 3.3 Categorical Categorical) has led to the noticeable sharp rise in pricing.

3.2 Categorical quantitative -

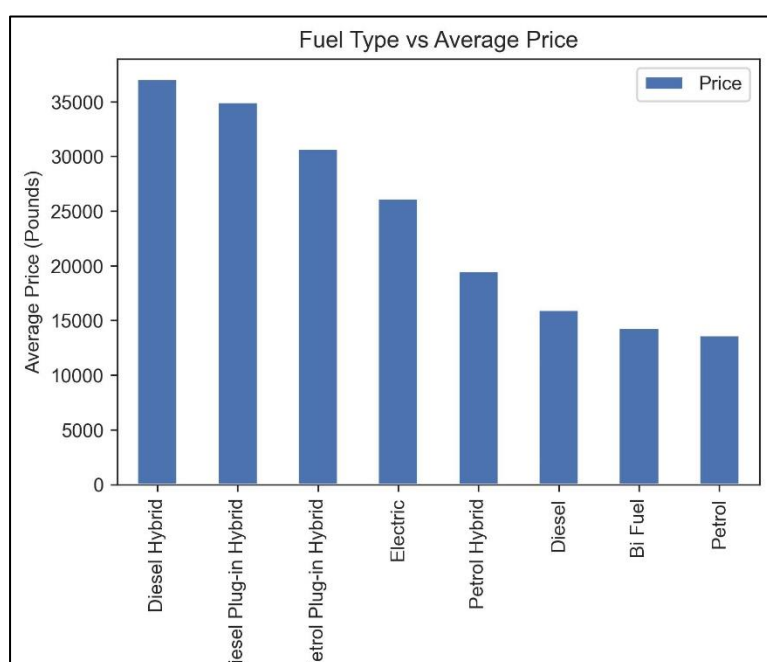


Figure 17 – Average Price of Cars Per Fuel Type.

Diesel cars are often more expensive to purchase and maintain than petrol equivalents because their engines need extra hardware, such as the selective catalytic reduction (SCR) system, which turns hazardous pollutants into benign ones. This holds true for all comparable fuel types that contain both diesel and petrol.

Cars with bi-fuels, which are the second least expensive, can run on a combination of natural gas and either diesel or petrol. However, the infrastructure in the UK for cars running on natural gas is not widely established, with few

petrol stations providing it so using it as an exclusive/primary means of fuel would prove challenging. Thus, rational inference dictates that after accounting for the difference in manufacture or conversion costs for cars to run on two different fuels, the final car price should be comparable to that of the car that is powered by one fuel in the UK. Though natural gas is less expensive per litre than diesel and petrol, the inaccessibility of the alternative fuel type means it has little bearing on the overall selling price. Running costs, however, should be cheaper if a consistent supplier can be used.

Unexpectedly, petrol hybrid vehicles are less expensive than pure electric cars despite implementing a combination of both electric and petrol fuel type technologies that complement each other. This is due to hybrid vehicles having smaller batteries than electric cars, a factor that significantly influences cost. Because of the larger, more costly battery and more intricate driving system, plug-in hybrids come at an even higher premium, with diesel hybrids being the priciest fuel type because, in addition to the diesel engine's added complexity, a sophisticated electric motor has been added and the cost of combining the two is significant.

Diesel engines are more efficient than petrol, however as both diesel and electric motors are great at producing torque at lower speeds, when paired together there is actually less efficiency. So creating a hybrid system, which improves efficiency but adds cost, on the top of a diesel engine, which also improves efficiency but also adds cost, as well as the lack of complementation between the two is currently illogical from a mass production perspective when competing with petrol hybrids and would require significant technical effort to achieve similar levels in performance at greater velocities. Nevertheless, they provide a distinctive driving experience, which is why there are 185 instances present in the data (6159 for petrol plug in for reference).

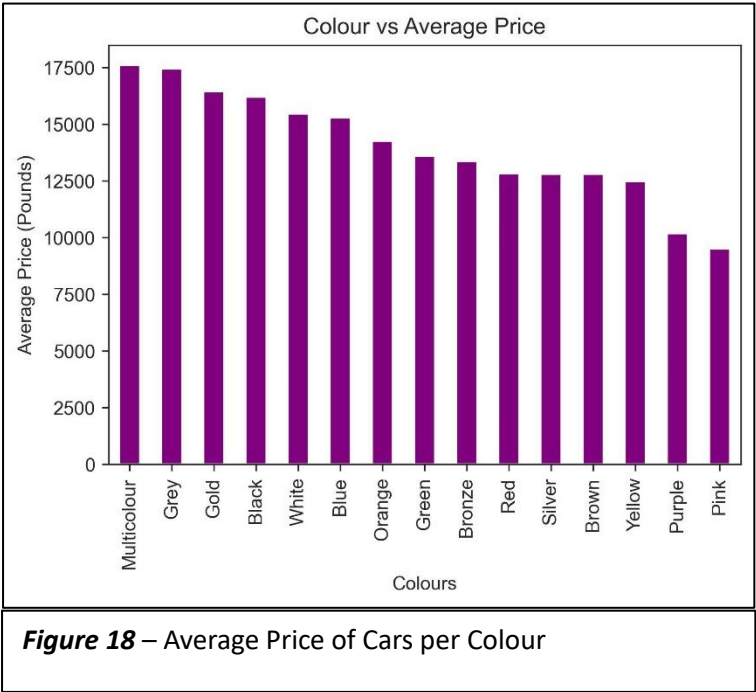
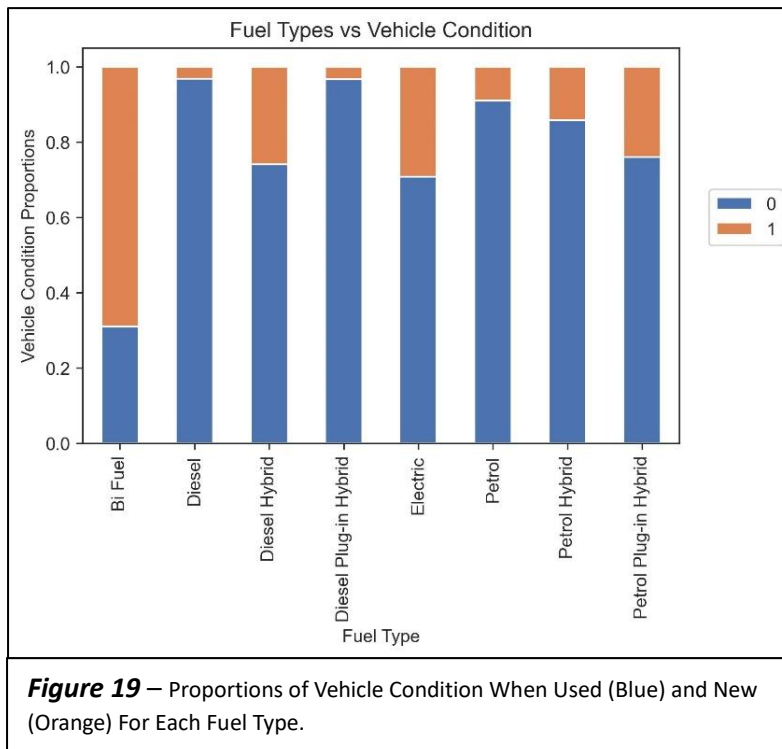


Figure 18 – Average Price of Cars per Colour

On average, pink and purple cars are the cheapest out of all the possible colours in the dataset. This appears to be a consequence of market segmentation and may bode to indicate the prevalence of cultural norms that uphold and reinforce gender stereotypes, in this case pink being a feminine colour exclusively for women and girls. Pink cars must not retain as much value in comparison with other colours due to the skew in dataset's distribution towards used cars, in combination with

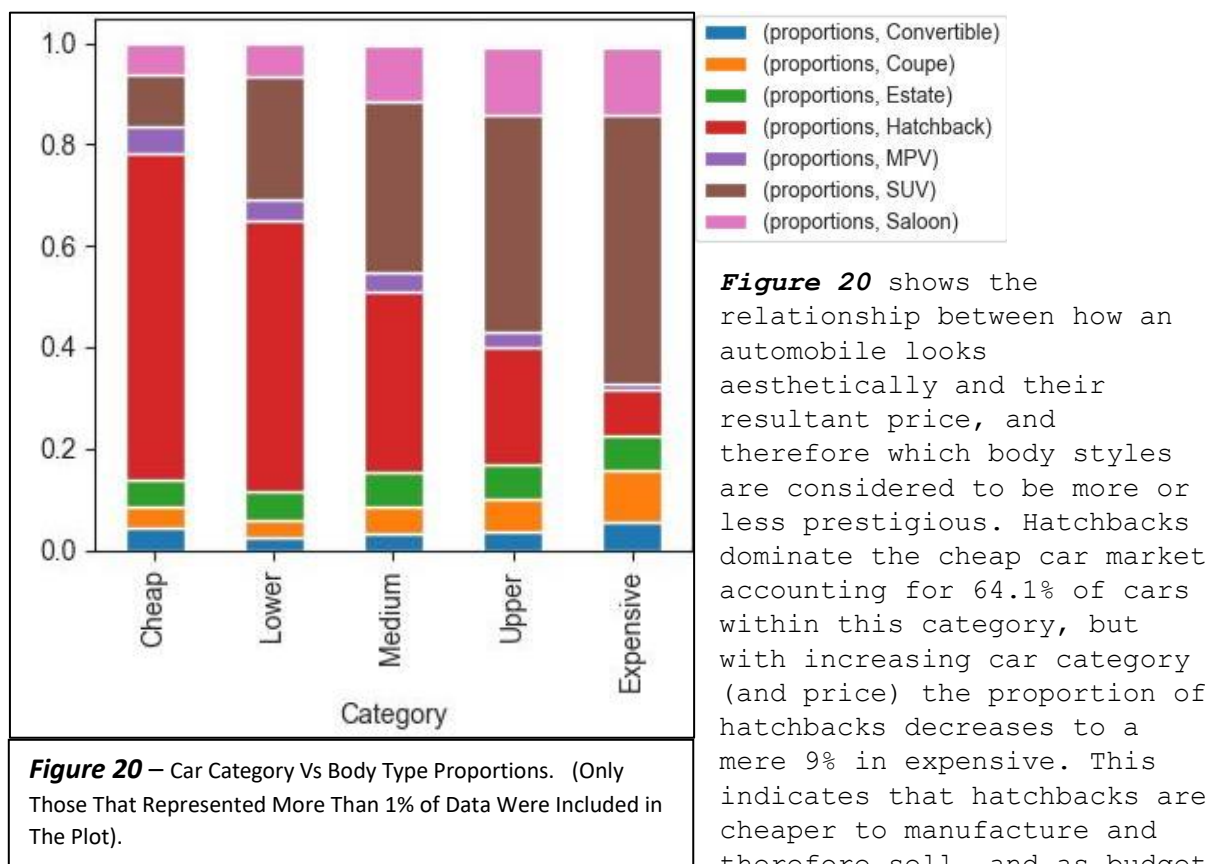
the limited market appeal towards men. Furthermore, the more costly models may not be produced in pink in similar proportions as other colours, while this doesn't hold true for the cheaper cars. In turn this may point to the larger issue of wage disparity between men and women. With respect to the most expensive, this is observed in multicolour. While it may be worth the added expense for cars on the higher end of the market to undergo the required paintwork expertise to produce a multicoloured vehicle, it becomes more illogical when paintwork contributes to a greater proportion to the overall price of the car, explaining the observed trend.

3.3 Categorical Categorical



The UK government first announced in 2017 that the sale of new petrol and diesel vehicles would be outlawed in 2035 as part of an initiative to decarbonise vehicles and reduce pollution. Petrol and diesel are the primary fuel types most widely used in internal combustion engines, however these are non-renewable and produce greenhouse gas emissions when burned which contribute significantly to climate change and global warming, as well as significantly reducing air quality. While electric vehicles indirectly contribute

to pollution through energy generation (assuming the electricity is generated using fossil fuels), they emit no tailpipe emissions, or other hazardous pollutants emitted by petrol and diesel. **Figure 19** displays visually that 91.0%, 96.8% and 96.7% of petrol, diesel and diesel plug in hybrid cars are used, while electric vehicles account for the second highest percentage (29%) of new cars. In combination with the diminishing new car market for petrol and diesel fuel types in line with meeting regulations by 2035, **Figure 19** helps describes the growing awareness of environmental concerns and the shift in the car market towards more sustainable and greener alternatives as electric vehicles are the key technology to decarbonise road transport. Comparing the plug in hybrids, 3.3% of diesel are new while that figure stands at 24% for the petrol equivalent, conveying the disparity in manufacturer rates for reasons detailed in 3.2 *categorical quantitative*. Bi-fuel is the only example where there are more new cars than used, which indicates the inability to sell these cars. This provides further strength to the fact that the infrastructure simply isn't in place for natural gas to become a widely used fuel type in the UK.



increases they appear to appeal less to consumers exemplified by the little representation in comparison as described.

The opposite is true for SUV's, with little prevalence in the cheap car market (10.1%) and increasing with increasing car category to become the majority body type for cars categorised as expensive (52.9%). Owning a SUV may be regarded as prestigious due to size, perceived safety and the association with luxury therefore appealing more to the consumer with a larger budget.

Convertibles account for a small proportion of cars in the dataset, attributed mostly to the weather conditions in the UK and lack of opportunity to utilize the open-air driving experience a convertible provides. Historically they have been seen as symbols of status and wealth often designed with performance in mind justifying why they are most prevalent in expensive, while noticeably reduced in upper, middle and lower categories. For the cheap category however, proportions comparable to that in expensive are observed, going against the expected grain. This may be a result of allowing a wider range of consumers who may not typically be able to afford the more expensive cars experience open top driving.

The relative proportions of estate cars remained virtually constant amongst differing categories. Estate cars tend to be longer versions of the car models which they're based, but not entirely individual models of themselves. The demand for extended versions of car models therefore remains constant irrespective of budget.